

Geometric Postulates, Theorems, and Proofs

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Geometry is built from a few **undefined**
terms: point, line, and plane.

Defined Terms

Defined terms are described using undefined terms.

Examples include **line segment**, **ray**, **angle**, and **collinear points**.

Postulate

A **postulate** (or axiom) is a statement accepted as true without proof.

Theorem

A **theorem** is a statement that must be proven using definitions, postulates, and previously proven theorems.

Proof

A **proof** is a logical argument that shows why a mathematical statement is true.

Common Postulates

Through any two distinct points, there is exactly one line.

A B

Exactly one line passes through A and B .

Common Postulates

If two lines intersect, they intersect at exactly one point.

$$\ell_1 \cap \ell_2 = \{P\}$$

Common Postulates

Through any three non-collinear points, there is exactly one plane.

A, B, C

determine exactly one plane.

Examples

Example 1

Identify each as an **undefined term**, **defined term**, **postulate**, or **theorem**.

Point

Ray

Through any two distinct points there is exactly one line.

Vertical angles are congruent.

Point: Undefined

Ray: Defined

Example 1

Two points determine one line: Postulate

Vertical angles are congruent: Theorem

Example 2

Classify each as **true** or **false**.

A line has two endpoints.

A plane extends infinitely.

A point has length and width.

A theorem must be proven.

False, True, False, True

Example 3

Complete the statements.

A line is determined by exactly _____ points.

A ray has one _____.

Three non-collinear points determine one _____.

two, endpoint, plane

Example 4

Name the defined term described.
Part of a line with two endpoints.

Line segment

Part of a line with one endpoint.

Ray

Example 4

Two rays sharing a common endpoint.

Angle

Example 5

Determine whether the statements are postulates or theorems.

Through any two points there is exactly one line.

Postulate

If two angles are supplementary to the same angle, then they are congruent.

Theorem

Example 5

Vertical angles are congruent.

Theorem

Example 6

State the reason for each statement.

Two points determine exactly one line.

Postulate

Vertical angles are congruent.

Theorem

Example 6

Three non-collinear points determine one plane.

Postulate

Example 7

Identify the undefined terms used to define each.

Line segment

Ray

Angle

Point, line, plane

Example 8

Arrange the proof steps in the correct order.

Apply the theorem.

State the given.

Write the conclusion.

Use the appropriate postulate.

2, 4, 1, 3

Practice Exercises

Practice 1

Identify each as an **undefined term**, **defined term**, **postulate**, or **theorem**.

Point

Ray

Through any two distinct points there is exactly one line.

Vertical angles are congruent.

Practice 2

Classify each as **true** or **false**.

A line has two endpoints.

A plane extends infinitely.

A point has length and width.

A theorem must be proven.

Practice 3

Complete the statements.

A line is determined by exactly _____ points.

A ray has one _____.

Three non-collinear points determine one _____.

Practice 4

Name the defined term described.

Part of a line with two endpoints.

Part of a line with one endpoint.

Two rays sharing a common endpoint.

Practice 5

Determine whether the following statements are postulates or theorems.

Through any two points there is exactly one line.

If two angles are supplementary to the same angle, then they are congruent.

Vertical angles are congruent.

Practice 6

State the reason for each statement.

Two points determine exactly one line.

Vertical angles are congruent.

Three non-collinear points determine one plane.

Practice 7

Identify the undefined terms used to define each.

Line segment

Ray

Angle

Practice 8

Arrange the proof steps in the correct order.

Apply the theorem.

State the given.

Write the conclusion.

Use the appropriate postulate.

Activities

Activity 1

Identify each as an **undefined term**, **defined term**, **postulate**, or **theorem**.

Plane

Line segment

If two lines intersect, they intersect at exactly one point.

The sum of the angles in a triangle is 180° .

Activity 2

Classify each as **true** or **false**.

A ray has two endpoints.

A line extends infinitely.

A plane has thickness.

Postulates are accepted without proof.

Activity 3

Complete the statements.

A ray has one _____.

A line extends infinitely in _____ directions.

Two distinct points determine one _____.

Activity 4

Name the defined term described.

A flat surface extending infinitely.

An exact location.

A straight path extending infinitely in opposite directions.

Activity 5

Determine whether the following statements are postulates or theorems.

Through three non-collinear points there is exactly one plane.

The base angles of an isosceles triangle are congruent.

If two lines intersect, they intersect at one point.

Activity 6

State the reason for each statement.

Three non-collinear points determine one plane.

Base angles of an isosceles triangle are congruent.

Two intersecting lines meet at exactly one point.

Activity 7

Identify the undefined terms used to define each.

Plane

Line

Angle

Thank you!